Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/07/2006
PI Name:	Matkowsky, Bernard Ph.D.		
Project Title:	FILTRATION COMBUSTION FOR MICROGRAVITY APPLICATIONS: (1) SMOLDERING AND FIRE SAFETY (2) COMBUSTION SYNTHESIS OF ADVANCED MATERIALS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-3107	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/04/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Gokoglu, Suleyman A	**Contact Phone:**	216-433-5499
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA18G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The task involves (1) proposing/developing theoretical models describing the fundamental mechanisms for the phenomena under consideration, (2) performing complementary analysis and numerical work on the proposed models and mechanisms, and (3) comparing the results of these analyses to experiments. The objective is to investigate combustion in porous media with applications to (1) smoldering and fire safety, and (2) combustion synthesis of advanced materials, also referred to as self-propagating high-temperature synthesis (SHS). Both smoldering and combustion synthesis of advanced materials (SHS) contribute to fundamental science, and also represent important applications for microgravity combustion science; in smoldering to fire safety, and in SHS to the determination of optimal synthesis conditions in both normal and microgravity environments. Combustion synthesis appears to compete favorably with conventional technology, by achieving shorter synthesis times, and at lower cost, by employing the		

	internal energy of the combustion reactions rather than the costly external energyof a furnace, employed in conventional technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A paper entitled, "On the transition from smoldering to flaming", coauthored by Profs. Matkowsky and Bayliss of Northwestern University and Dr. Anatoly Aldushin of the Russian Academy of Science's Insatitute for Structutal Macrokinetics and Materials Science, was published (Combustion and Flame, vol. 145 (2006), 579-606). In addition, a version of the paper, entitled "On the mechanism of triggering the transition from smoldering to flaming" was presented at the 31st International Symposium on Combustion, held in Heidelberg, germany in August, 2006. It was accepted for publication in the Symposium Proceedings. The abstract for both papers reads: "We consider the transition from smoldering to flaming. Though there have been numerous experimental studies of this topic, it has been virtually untouched theoretically. Rather than investigating the details of either the transition process or the subsequent flaming, we focus on determining the mechanism and the conditions that trigger the transition. We employ both computations and approximate analytical approaches. We consider a planar, forward smolder wave driven by a constant forced flow of gas containing oxidizer. The chemical kinetic scheme employed consists of three reactions, namely, pyrolysis, fuel oxidation and char oxidation. There have been a number of speculations about the nature of the triggering mechanism, including the gaseous reactions, the char oxidation reaction, destruction of the porous matrix through which the smolder wave propagates, and others. However, no mechanism has, as yet, been theoretically demonstrated to be capable of acting as the triggering mechanism. We show that the char oxidation reaction hardly affects the characteristics of smolder wave propagation due to its small reaction rate, though under appropriate conditions, it can act as the trigger for the transition to flaming due to its ability to self-accelerate. Specifically, we introduce the concept of, and then compute, a quantity which we term the flaming distance L(F) . This is the distance that a steadily propagating smolder wave initiated at the gas flux inlet, travels inside the porous medium before the char oxidation reaction spontaneously self-accelerates, resulting in an eruption of the temperature in the smolder front. That is, the smolder wave propagates for a relatively long latent period of time until it reaches L(F) . A transition to flaming then occurs. The flaming distance L(F) depends on the physicochemical parameters of the fuel and the products as well as external parameters, such as the velocity and composition of the incoming gas, heat loss, etc. We show that smolder waves propagating in porous samples of length L do (do not) exhibit a transition to flaming if L(F) < L (L(F) > L)). Finally, a new paper is currently in preparation. In contrast to the papers which consider forward smoldering, the new paper considers the possible transition from smoldering to flaming for reverse smolder waves.
Bibliography Type:	Description: (Last Updated: 11/08/2006)
Articles in Peer-reviewed Journals	Aldushin AP, Bayliss A, Matkowsky BJ. "On the transition from smoldering to flaming." Combustion and Flame, 2006 May;145(3):579-606. , May-2006
Papers from Meeting Proceedings	Matkowsky BJ, Bayliss A, Aldushin AP. "On the mechanism of triggering the transition from smoldering to flaming." 31st International Symposium on Combustion, held in Heidelberg, Germany, August, 2006. 31st International Symposium on Combustion. Proceedings. August 2006. , Aug-2006